Exploring the Depths of 3D Semantic Novelty Detection

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Abstract

In this project we will focus on 3D semantic novelty detection. This task involves identifying whether a given data sample belongs to a predefined nominal distribution. In our case, the nominal data distribution is composed of data samples categorized into a set of known semantic classes. For each test sample, our goal is to determine whether it belongs to one of the known semantic classes constituting the nominal data distribution (in-distribution) or if it does not (out-of-distribution). We will rely on the 3DOS benchmark for the 3D semantic novelty detection task definition and we will experiment on the Synthetic to Real scenario. In this particular scenario the nominal data samples are Synthetic (point cloud sampled from CAD models) and test samples are real-world point clouds obtained through sensor scan.

1. Introduction

Many machine learning models assume that training and testing data are drawn independently and identically distributed (i.i.d.), a reasonable assumption for controlled lab settings. However, this assumption often fails in real-world deployments, where various shifts in distribution relative to the training data can occur. These shifts may include the introduction of new object categories leading to semantic changes or data from different domains causing covariate shifts. Such shifts can occur individually or concurrently, and samples that deviate from the training data are typically referred to as out-of-distribution (OOD) data. These OOD data pose significant risks for autonomous systems, as evidenced by numerous incidents involving self-driving cars encountering unfamiliar objects on the streets.

Taking inspiration from the 3DOS [1] paper, our project goes through the following steps.

Here is the link to the project's GitHub repository

1.1. Semantic novelty detection baselines

Evaluate some simple baselines from the family of "Discriminative Methods" and "Representation and Distance Based Methods".

1.2. Semantic novelty detection analysis

Document and examine the outcomes achieved in Sec. 1.1. Additionally, we will perform a failure-case analysis for two methods from each of the families. The analysis will aim to identify the challenges encountered by these methods and explore the reasons behind them. Furthermore, we will explore strategies to enhance method performance.

1.3. Fine-Tuning free semantic novelty detection

Investigate the incorporation of large-scale pre-trained models (we chose OpenShape [10]) to improve the task of 3D semantic novelty detection. Rather than training a specialized model for each set of nominal data samples, the approach involves leveraging a general purpose pre-trained model to extract feature representations of both nominal and test data distributions. For computing normality score will be employed methods from the family of "Representation and Distance Based Methods".

1.4. Extension Point: Real World application

Suggest and assess practical applications for 3D semantic novelty detection, including but not limited to fields like robotics, medical imaging, and augmented reality. Additionally, evaluate the integration of the 3D semantic novelty detection method within one of these applications. In real-world scenarios, 3D scans are often incomplete, cluttered with noise, and affected by scan artifacts.

2. Related Works

We provide an overview of the architectures we exploited to develop the project and the benchmarks we used to measure experiments' results.

2.1. PointNet++

PointNet++ [2] introduces a hierarchical neural network that applies PointNet [3] recursively on a nested partitioning of the input point set. By exploiting metric space distances, the network is able to learn local features with increasing contextual scales. Further more, the architecture incorporates novel set learning layers to adaptively combine features from multiple scales.

2.2. DGCNN

Dynamic Graph CNN [16] proposes a new neural network module dubbed EdgeConv suitable for CNN-based high-level tasks on point clouds with the purpose to recover topology that can enrich the representation power of point clouds. EdgeConv acts on graphs dynamically computed in each layer of the network. It incorporates local neighborhood information, can be stacked applied to learn global shape properties and is able to capture semantic characteristics in feature space, over potentially long distances in the original embedding.

2.3. 3DOS: Synthetic to Real Benchmark

The goal of OOD detection is to identify and reject samples with any distribution shift with respect to the training data. In the particular case of semantic novelty detection, the concept of novelty is limited to the categories not seen during training, regardless of the specific domain appearance of the observed samples.

The 3DOS paper defines benchmarks to understand the quality of a model to perform semantic novelty detection. It introduces the Synthetic to Real-World cross-domain scenario: synthetic point clouds from ModelNet40 [18] for training while testing on real-world point clouds from ScanObjectNN [8].

To understand the ability of the network to detect OOD samples two main metrics are used: AUROC and FPR95 [4]. On the other hand, for ID samples the accuracy (ACC) is exploited to evaluate the correctness of the predicted class.

The final output of the network should be the confidence about the distribution of the samples. To address this fact, discriminative and distance based baselines are employed at the end of the network, in particular MSP [5] and MLS [12] as discriminative and L^2 norm [11] and SubArcface [6] as distance based.

2.4. OpenShape

OpenShape [10] is a method for learning multi-modal joint representations of text, images, and point clouds adopting the commonly used multi-modal contrastive learning framework for representation alignment, but with a specific focus on scaling up 3D representations to enable open-world 3D shape understanding.

3. Methodology

3.1. Backbones

Semantic novelty detection task performance has strong dependency from the architecture used as a backbone of the network, so it is important to explore different solutions like PointNet++ and DGCNN and compare them. We expect slightly different results as one exploits the geometry position of the points and the other exploits semantic meaning of them. In contrast of these standard approaches, we employ OpenShape as a multi-modal approach, used to dealing also with different representations and trained on that types. In this way the features extracted should have different meaning and incorporate more wide background.

3.2. Baselines

3.2.1 Discriminative Methods

By training a model with multi-class supervision we expect to get low uncertainty on in-distribution (ID) data and high uncertainty for OOD samples. Thus, a baseline approach may consist in using the maximum softmax probability (MSP) [5] as a normality score to separate known and unknown instances. Another very recent work has discussed how the normalized softmax probabilities can be replaced by the maximum logit scores (MLS) [12], resulting in an approach competitive with other more complex strategies. A key part of this implementation is that all the methods of this discriminative family are applied post-hoc on the closed set classifier, meaning that the original training procedure and objective are not modified. Thus, the models maintain their ability to distinguish among the known classes.

3.2.2 Representation and Distance Based Methods

Enhancing data representation may help to better characterize known data and consequently make the identification of unknown samples easier. In a reliable embedding space, OOD samples should be far away from ID classes so that the distance from stored exemplars or prototypes can be used as a scoring function. Some solutions build on discriminative models but, instead of considering the prediction output, they focus on the learned features and evaluate sample distances by using different metrics like L^2 norm [11].

The SubArcFace [6] approach belongs to the family of margin-based softmax methods that aim at simultaneously achieving maximal intra-class compactness and inter-class discrepancy without the drawbacks (negative sampling and large data batches) that affect the triplet and the contrastive losses. With respect to other similar strategies, for each known class, SubArcFace identifies multiple sub-centers, and a training sample only needs to be close to one of them rather than to a single class prototype. It should be noted that CE(L^2) and SubArcface require training data to be available also at test time since they compute the test sample normality score as distance to the nearest train sample.

Figure 1. Dataset subdivision

3.3. Dataset

ModelNet40 [18] is chosen because it has a better overlap with ScanObjectNN [8] and previous works already considered the same cross-domain scenario in the context of point cloud object classification. We define three different category sets: SR1, SR2, and SR3 as described in 3. The first two sets are composed by matching classes of ModelNet40 and ScanObjectNN. The third set (SR3) is instead composed by ScanObjectNN classes without such a one-to-one mapping with ModelNet40. Overall we have two scenarios with either SR1 or SR2 used as known and the other two considered as unknown. For the Synthetic-To-Real Scenario, models are trained on ModelNet40 samples of the known classes and evaluated on the ScanObjectNN samples of both known and unknown classes.

(a) ModelNet40 chair sample (b) ScanObjectNN chair sample

3.4. Metrics

AUROC (Area Under the Receiver Operating Characteristic Curve) [4] is a threshold-free metric used to evaluate binary classification models. It represents the probability that a randomly chosen positive sample will have a higher predicted probability than a randomly chosen negative sample.

FPR95 (False Positive Rate at 95% True Positive Rate) [4] is a metric that measures specificity at a sensitivity level of 95%. It involves selecting a threshold ensuring 95% of positive samples are correctly identified and then computing the false positive rate. Lower FPR95 values indicate better performance in limiting false positives while maintaining high sensitivity.

3.5. Failure-case analysis

To analyze the reasons behind of the failing predictions in OOD task we explore the distribution of the misclassified samples related to their effective classes and their predicted classes. In this way we aim at understanding which OOD objects are geometrically and semantically similar to ID objects, furthermore we should be able to detect which class the network is uncertain of, resulting in low confidence that corresponds to low capability to extract meaningful information during training.

In order to explore the problem we need to fix a threshold to be able to target misclassified objects specifically for that threshold, because all the previous metrics are threshold free or have intrinsic value in their definitions. We select two possible thresholds that maximise two criteria:

F1 score [17]:

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Youden's index J [13]:

$$J = TPR - FPR$$

According to that, all the values above the threshold are classified as ID and all the values below as OOD. For each sample we investigate the confidence value, its real class and the predicted one.

Further more, we employed a possible technique to improve the weaknesses relative to the domain shift about the synthetic to real gap. The solution consists in emulating the corruptions observed in ScanObjectNN dataset at training time by adopting the Occlusion and LIDAR augmentations [7] of the ModelNet40 objects.

4. Experiments

The goal of the project was to evaluate some baselines from the family of "Discriminative Methods" and "Representation and Distance Based Methods". In particular, we perform a failure-case analysis on the semantic novelty detection task over the two backbones (PointNet++ and DGCNN) to identify what the limit cases of these two methods are and how to enhance their performance, exploiting training over occluded pointclouds for instance. Finally, we took advantage of some pretrained models of OpenShape in order to compare the results of not-specifically-trained models like OpenShape ones and our custom-trained models using DGCNN or PointNet++ as backbones.

4.1. Backbone and baselines comparison

Interestingly, PointNet++ backbone seems to be more robust to the domain shift than DGCNN. How shown in Tab. 3 the difference between the networks' performances are significantly better for most of the baselines in favor of PointNet++.

Discriminative methods have a relevant boost of performance using PointNet++ architectures on SR1. Probably because the simplicity (looks only for geometrical relationship between points) of the network matches almost perfectly with the simplicity of the dataset, so it can be more effective without useless components and operations.

DGCNN [16]				
	SR1 (easy)		SR2 (hard)	
Method	AUROC	FPR95	AUROC	FPR95
MSP	71.92	91.50	61.59	88.25
MLS	69.73	91.68	61.82	87.25
CE(L^2)	68.57	**85.99**	66.34	88.15
SubArcFace	**73.38**	**85.99**	**69.27**	**80.26**

Table 1. Results of DGCNN backbone on Synthetic to Real Benchmark track. Trained on the specified subsets.

PointNet++ [2]				
	SR1 (easy)		SR2 (hard)	
Method	AUROC	FPR95	AUROC	FPR95
MSP	82.80	79.76	67.62	88.49
MLS	**84.47**	**72.42**	65.66	87.11
CE(L^2)	81.74	80.61	70.74	**79.54**
SubArcFace	77.72	88.07	**72.87**	85.01

Table 2. Results of PointNet++ backbone on Synthetic to Real Benchmark track. Trained on the specified subsets.

PointNet++ vs DGCNN				
	SR1 (easy)		SR2 (hard)	
Method	AUROC	FPR95	AUROC	FPR95
MSP	10.88	21.74	6.03	-0.24
MLS	14.74	19.26	3.84	0.14
CE(L^2)	13.17	5.38	4.40	8.59
SubArcFace	4.34	-2.08	3.6	-4.75

Table 3. Difference between PointNet++ and DGCNN results in term of percentage points. Green color in favor of PointNet++ and red in favor of DGCNN.

The results of distance based methods are highly dependent on the specific backbone chosen. In particular, we can observe that SubArcFace method outperforms other methods in DGCNN but slightly under performs in PointNet++. This can be explained by looking at which information is extracted by the networks: DGCNN enriches the feature space with semantic meaning, instead of only considering the geometrical ones.

4.2. Failure-case Analysis

We performed failure-case analysis on MSP [5], MLS [12], CE(L^2) [11] and SubArcFace [6] methods on both backbones, on both training datasets split (SR1 and SR2), exploiting F1 [17] and J [13] thresholds.

4.2.1 Threshold evaluation

We briefly discuss the two metrics to chose the threshold inside failure-case analysis. We evaluate them by comparing the accuracy in Semantic Novelty task relative to a binary classifier with threshold set accordingly to F1 and J.

DGCNN				
	SR1 (easy)		SR2 (hard)	
Method	F1 ACC	J ACC	F1 ACC	J ACC
MSP	68.20	**71.38**	54.15	**55.78**
MLS	44.91	**70.80**	**72.66**	55.40
CE(L^2)	58.44	**64.67**	54.95	**63.46**
SubArcFace	68.37	**70.59**	**72.56**	47.23

Table 4. Results of accuracy using thresholds from F1 and J on DGCNN backbone on Semantic Novelty task.

PointNet++				
	SR1 (easy)		SR2 (hard)	
Method	F1 ACC	J ACC	F1 ACC	J ACC
MSP	76.57	**78.24**	62.49	**67.16**
MLS	79.31	79.31	55.50	**62.32**
CE(L^2)	56.54	**77.47**	70.74	**79.54**
SubArcFace	75.05	**75.92**	67.58	67.58

Table 5. Results of accuracy using thresholds from F1 and J on PointNet++ backbone on Semantic Novelty task.

As Tab. 4 and Tab. 5 show, J metric provides overall better results. This usually results in a better partition between ID and OOD misclassified samples trying to minimize this values. Even if the accuracy is not so specific and reliable as an evaluation metric, it gives enough information to intuit which threshold metric is more indicated in this context. To slightly better understand we investigate also FP and FN, looking at their balance. As an example, the best accuracy of 79.54 with configuration of PointNet++ and CE(L^2) on SR2 with J metric generates 273/788 FP and 664/2102 FN. When looking at two limit cases as the one in which F1 performs better, we can show that this is an unbalanced situation. In fact, the first generates 788/788 FP and 3/2102 FN and the second 788/788 FP and 5/2102 FN.

Further more, we reiterate the concept that PointNet++ has better results than DGCNN. To make the paper easier to be read, we chose to only report the results on J threshold on PointNet++, being that the results using F1 threshold and DGCNN architecture bring to similar considerations. For completeness, the results not directly discussed in the paper are reported in the GitHub repository**??**.

4.2.2 PointNet++

We report the main results of our failure-case analysis on different methods using PointNet++ as backbone. The following tables have on their columns, in order: the true class

of the misclassified samples, the wrong prediction, the number of times the wrong predictions were made, the closest classes to which the network would have assigned the samples if it was a classification task and then the max, min and average scores the samples were assigned.

SR1-MSP						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
sofa	OOD	75	sofa	0.9056	0.5181	0.9968
shelf	OOD	69	shelf	0.9099	0.4387	0.9969
sofa	OOD	50	chair	0.8672	0.4410	0.9968
cabinet	ID	34	door	0.9996	0.9982	1.0000
desk	ID	27	chair	0.9991	0.9976	1.0000
display	ID	19	shelf	0.9988	0.9978	0.9999

Table 6. Acc: 0.7824, Th: 0.9970

SR1-MLS						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
shelf	OOD	81	shelf	2.9954	0.2377	4.2484
sofa	OOD	58	sofa	3.0268	0.5446	4.2039
sink	OOD	46	sink	2.6251	0.4826	4.0969
cabinet	ID	35	door	6.2750	4.7474	8.3745
desk	ID	35	chair	5.4413	4.3501	8.4033
toilet	ID	25	chair	5.5528	4.3188	7.5376

Table 7. Acc: 0.7931, Th: 4.3004

We can see that the two Discriminative Methods bring similar but not equal results. In fact, the three major errors on OOD in MLS are caused by the fact that the model is not sure enough of the predictions it is giving to exceed the set threshold, although they are right. On the other hand MSP, struggles to distinguish between sofas and chairs.Both methods confuse desk with chairs and cabinets with doors.

SR1-CE(L^2)						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
sofa	OOD	165	shelf	0.0998	0.0701	0.1230
shelf	OOD	103	shelf	0.0995	0.0736	0.1228
sink	OOD	100	shelf	0.0891	0.0681	0.1217
cabinet	ID	67	shelf	0.2401	0.1240	0.7613
desk	ID	45	shelf	0.1579	0.1248	0.2623
display	ID	44	shelf	0.1543	0.1249	0.2206

Table 8. Acc: 0.7491, Th: 0.1240

As it can be observed from the results, PointNet++ struggles when exploiting Representation and Distance Based Methods as it is not capable of understanding semantic distances between samples but it only uses geometric based distances. As a result, exploiting CE(L^2) method results in not understanding the shelf labelled samples while in SubArcFace it is not sure enough of the predicted labels and cannot distinguish displays from shelves, cabinets from doors and toilets from chairs.

SR1-SUBARCFACE						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
chair	OOD	102	chair	0.9820	0.8782	0.9995
shelf	OOD	72	shelf	0.9757	0.8828	0.9995
sofa	OOD	63	sofa	0.9742	0.8526	0.9995
display	ID	33	shelf	0.9998	0.9996	1.0000
cabinet	ID	31	door	0.9999	0.9996	1.0000
toilet	ID	16	chair	0.9998	0.9996	1.0000

Table 9. Acc: 0.7592, Th: 0.9995

If we look at the accuracy obtained with both Discriminative and Distance Based Methods, it's clear that training our backbone on SR2 (defined as hard) obtains worse results in Semantic Novelty Detection w.r.t training on SR1.

SR2 - MSP						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
desk	OOD	151	desk	0.9197	0.4791	0.9987
desk	OOD	41	bed	0.8008	0.4524	0.9979
bed	OOD	33	desk	0.8994	0.5010	0.9988
door	ID	218	display	0.9999	0.9978	1.0000
cabinet	ID	107	display	0.9998	0.9958	1.0000
shelf	ID	45	display	0.9997	0.9958	1.0000

Table 10. Acc: 0.6716, Th: 0.9988

SR2 - MLS						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
desk	OOD	134	desk	2.4596	0.1589	3.9161
desk	OOD	37	bed	1.6767	-0.3290	3.8505
bed	OOD	29	desk	2.5544	0.7649	3.8242
door	ID	221	display	6.5408	4.3508	8.1029
cabinet	ID	140	display	5.7606	4.0167	7.6563
shelf	ID	66	display	5.3616	4.0454	7.4328

Table 11. Acc: 0.6232, Th: 3.9766

It is pretty clear that the model fails at distinguishing samples belonging to "desk" and "bed" labels. This behaviour can be encountered exploiting both MSP and MLS methods. As a result, even if the model assigns a sample of these classes to the correct label, it does not have much

confidence about its decision, so it does not give a score that allows it to go over the set threshold.

On the other hand, when classifying as ID while being OOD, both methods fail by assigning samples belonging to "door", "cabinet" and "shelf" classes to the label "display".

SR2 - CE (L^2)						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
desk	OOD	144	bed	0.0905	0.0651	0.1145
toilet	OOD	50	bed	0.0898	0.0661	0.1133
bed	OOD	48	bed	0.0933	0.0732	0.1147
door	ID	214	bed	0.1973	0.1161	0.5596
cabinet	ID	179	bed	0.1778	0.1154	0.8518
shelf	ID	102	bed	0.1726	0.1161	0.4304

Table 12. Acc: 0.6540, Th: 0.1149

SR2 - SubArcFace						
True class	Pred	Times	Closest class	Mean score	Min score	Max score
desk	OOD	107	desk	0.9789	0.8697	0.9993
desk	OOD	37	shelf	0.9726	0.8477	0.9993
display	OOD	32	chair	0.9886	0.9400	0.9993
sofa	ID	68	bed	0.9998	0.9995	1.0000
cabinet	ID	66	desk	0.9998	0.9994	1.0000
cabinet	ID	66	display	0.9998	0.9994	1.0000

Table 13. Acc: 0.7187, Th: 0.9993

Interestingly, CE(L^2) method provides the same results of SR1 in SR2. In fact, it demonstrates that this model is not capable of exploiting well the semantic differences among the samples. The only difference between the two trained models is that the closest label is always "bed" while in SR1 it is "shelf".

While using SubArcFace method, the same behaviour regarding "desk" label occurs in OOD misses, it seems that this class is pretty hard for our backbone to be understood. Another issue is represented by the "cabinet" samples that are often classified as ID and recognized as "desks" or "displays".

Figure 3. Desk sample from ScanObjectNN SR2

4.3. Methods Performance Enhancement

So far, we have trained our models with (synthetic) point clouds, which well represent objects in their canonical form. It is not wrong, but it does not fully represent reality: real-world point clouds are affected by partiality, occlusion, cluttered with noise and background points. In order to partially close this gap in the domain shift, a solution consists in emulating these corruptions conditions at training time.

We experimented with this solution by adopting the *Occlusion* and *LIDAR* augmentations from [7]. We performed experiments with DGCNN backbone, for MSP, MLS baselines and euclidean distance CE(L^2).

DGCNN with corruption				
	SR1 (easy)		SR2 (hard)	
Method	AUROC	FPR95	AUROC	FPR95
MSP	80.75	79.39	63.70	91.39
MLS	**81.32**	**70.69**	**62.66**	90.96
CE(L^2)	76.08	78.84	64.16	**84.25**

Table 14. Results of DGCNN backbone on Synthetic to Real Benchmark track. Trained on the specified subsets.

Corrupted images vs Standard images				
	SR1 (easy)		SR2 (hard)	
Method	AUROC	FPR95	AUROC	FPR95
MSP	8.8	12.11	2.11	-3.14
MLS	11.59	20.99	0.84	-3.71
CE(L^2)	7.51	7.15	-2.18	3.9

Table 15. DGCNN backbone. Result comparison between regular and corrupted images.

Tab. 15 shows how using corrupted images in the training phase improves the performance of our architecture, especially when trained on SR1. By using a training set that is more "realistic", more close to the test one, the domain shift results in having less impact: our backbone now results less over-fitted on synthetic regular objects and can generalize more.

4.3.1 Further possible method performance enhancement: pretraining with OcCo

We wanted to point out that another possible method performance enhancement could be performing pretraining of our DGCNN and PointNet++ backbones with OcCo [14]. This pretraining method basically consists in masking all points occluded in a camera view, learning an encoder-decoder model to reconstruct the occluded points and using the encoder weights as initialisation for downstream point cloud tasks. As the paper reports, this pretraining method leads to significant improvements in basically every downstream

task following it. We then assume that it could also lead to improvements in 3d semantic novelty detection task.

4.4. Pretrained models: OpenShape

The experiments taken on OpenShape are conducted following distance based methods. In fact, the network extracts from the point clouds the meaningful features and for each point finds the nearest: based on how much they are far from each other, we obtain a confidence score.

OpenShape CE(L^2)				
	SR1 (easy)		SR2 (hard)	
Variant	AUROC	FPR95	AUROC	FPR95
B32	81.79	65.75	65.61	93.01
G14	55.17	96.70	55.75	91.39
L14	81.77	69.91	59.72	90.87

Table 16. Results of Openshape pre-trained model on Synthetic to Real Benchmark track. Evaluate with euclidean distances in a non-normalized space.

OpenShape vs PointNet++ CE(L^2)				
	SR1 (easy)		SR2 (hard)	
Variant	AUROC	FPR95	AUROC	FPR95
B32	0.05	14.86	-5.13	-13.47
G14	-26.57	-16.09	-14.55	-11.85
L14	0.03	10.70	-11.02	-11.33

Table 17. Difference between Openshape and PointNet++. Green color in favor of Openshape and red in favor of PointNet++.

Usage of a pre-trained multi-modal model seems to have strong dependencies w.r.t training and test dataset. In fact, as the results show, performances change significantly between test on SR1 and SR2. The reasons behind this behaviour can be found in the intrinsic mechanism of the model: given that OpenShape outputs feature vectors, the distance of the test samples from the trained objects' representation is mainly related on the different categories and shapes of both datasets. So we can assume that SR1 configuration has higher discrepancy on OOD samples and lower with ID samples w.r.t. samples of training dataset than SR2.

Further more, we investigate on why there is significant difference of performance among the Openshape variants and we find three main reasons that explain the poor results of G14 in comparison to B32 and L14. The first is the training dataset of each variant: G14 is the only one to be trained on all the four datasets of OpenShape paper, including ModelNet40, instead the other two trained only on Objaverse [9]. For this reason G14 variant can have a different consideration of the tested objects. Secondly, each variant defines a gravity axe to align the object according to a specific directions (x, y, z) and this changes drastically its perception. In fact, G14 is the only one to represent point clouds on z-axe instead of y-axe and this increases the difficulty of the network to understand the object. Lastly, there is a difference about the architecture and the number of total parameters to be trained. G14 has significantly more parameters, about 32.3M of parameters w.r.t 13.3M of the other two in the PointBert [15] backbone. This leads to overfitting the network during training and to get a less generalized model at the end.

5. Extension Point: Real-World application

3D Semantic Novelty Detection could push the boundaries of many real world applications. In fact, being able to detect anomalies on real world objects scans having trained machine learning models over "perfect" 3d CAD models of the same objects could lead to major improvements in fields like robotics, medical imaging and augmented reality.

For example, it could be useful in assembly lines of every kind: there would be an initial phase where CAD models of the wanted objects would be created and used as training samples and then a sensor scan would be performed over all the objects that go through the assembly line. If an object is classified as OOD, a human operator would intervene and personally check it.

Another example, this time in medical imaging, could be training our models over computer-created scans of people mouths and then performing semantic novelty detection over real time scans of patients to discover whether there is a malformation of palates or some kind of dental pathology detected as OOD by the trained model.

Clearly, in both these scenarios, imperfections due to noise or incomplete scans would affect models performances. As shown in Sec. 4.3 performing occlusions also on training samples would probably partially positively affect their performance.

6. Conclusion

As a result of our analysis we can conclude that the performances of the 3D Semantic Novelty Detection methods depend on the chosen backbone: PointNet++ has proven to be extremely robust in processing real-world data across domains.

Secondly, we went in depth of our misclassification errors to discover why and how our models predicted those samples like that, giving us a better understanding of the models and their limits.

Moreover, we verified the importance of applying some method performance enhancement operations like training on occluded pointclouds to better generalize when going from Synthetic to Real scenarios. In fact, this methods led to a major improvement on the DGCNN backbone performances.

Finally, we evaluated the possibility of employing some multimodal pretrained models instead of training from the ground up a specific model and we also introduced some possible real world applications of the 3D Semantic Novelty Detection task.

References

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